

Moustafa Yehia Hassan

AI / NLP Researcher · LLM Evaluation · Model Auditing · Trustworthy NLP

Doha, Qatar · +974 77701880 · moustafayhassan@gmail.com

[LinkedIn](#) · [GitHub](#) · [Reproducibility Repository](#) · escapingplanetearth.blogspot.com

RESEARCH PROFILE

Interdisciplinary AI/NLP researcher specializing in trustworthy LLM evaluation, distributional-fidelity auditing of synthetic clinical text, computational mental-health NLP, and interpretability-oriented probing experiments. My work asks when language models and NLP classifiers appear accurate while relying on brittle lexical shortcuts, annotation artifacts, stereotyped over-simulation, or population-level geometric distortion, and it builds statistically rigorous, reproducible frameworks that surface these failures before claims of clinical or benchmark validity. I pair graduate AI training and an M.A. in Comparative Literature with a decade of institutional experience in data governance, auditability, and reproducible reporting.

RESEARCH FOCUS

- LLM and synthetic-data evaluation; distributional-fidelity auditing; benchmark, persona, and calibration validity.
- Interpretability experiments: hidden-state probing, activation steering, and mode-aware generation over frozen LLMs.
- Computational mental-health NLP: shortcut-learning and label-bias diagnostics; atypicality-graded failure analysis.
- Biomedical NLP reliability: deterministic preprocessing, margin-based abstention, safety gates, and token-level audit trails.
- Reproducible ML: OSF pre-registration, bootstrap inference, leakage control, and full data-integrity verification.

PUBLICATIONS & PEER-REVIEWED WORK — ACCEPTED

The Divergence Hypothesis: Unmasking Lexical Interference and Label Bias in Mental Health NLP

Accepted — ACL BioNLP 2026

- Introduces **TSS (Triple-Stream Stress probe)**, a white-box framework decomposing text into lexical character n-grams, content-free morpho-syntactic structure, and psycholinguistic style; and **DoD (Degree of Divergence)**, a Difference-in-Differences metric adapted from econometrics to quantify annotation-regime bias.
- Across four datasets (N = 12,906) demonstrates a **Lexical Interference Paradox**: adding lexical features degrades Macro-F1 on human labels (~7.2 pp) but not on auto-labels; DoD = 0.0374 (95% CI [0.0097, 0.0651], Cohen's d = 2.63); interventional masking retains ~99% of structural performance after destroying all content words.
- Bootstrap inference (n = 10,000), BH-FDR correction, ElasticNet probes, nested leave-one-domain-out evaluation; open-source reproducibility repository (MIT).

The Signal in the Noise: An Auditable Reliability Layer for Biomedical Text Classification

Accepted · IEEE ICBCB 2026 (to appear)

- Author / corresponding author. A conservative, deterministic spell-correction reliability layer (bounded edit-distance candidates, corpus-derived n-gram scoring, biomedical safety gates, margin-based abstention, token-level decision traces) built on a medical do-no-harm principle.
- Evaluated on a manually curated 2,104-token benchmark and a 10,000-example tri-class CORD-19 protocol: **94.61% error-fix recall, zero harmful edits on negative controls, ~80% recovery of noise-induced Macro-F1 degradation**; BioBERT case study with scispaCy/UMLS entity normalization and McNemar's exact test. Open-source repository (MIT).

MANUSCRIPTS — UNDER REVIEW / IN PREPARATION

The Stereotype-Lock Effect: Over-Simulation and Variance Compression in LLM Simulated-Patient Personas

Manuscript under review

- Pre-registered distributional-fidelity audit of 20 LoRA-fine-tuned depression personas against a clinical human reference using the windowed TSS-W framework; a directional rejection of the pre-registered hypothesis (AFI-

W = +0.147) with variance compressed to ~1/10 of the human reference — over-simulation rather than flattening.

The Population-Geometry Gap: Prompt-Conditioned Patient Simulation that Looks Right Sentence-by-Sentence but Deforms as a Population

Manuscript under review

- Introduces **TADA (Tri-Axial Diversity Audit)** and **FRFA (Feature-Relation Fingerprint Audit)** over five generator families; a convergent-failure argument shows two independent distributional baselines (MMD, MAUVE) judge the most geometrically deformed family among the closest to human — motivating geometry-level diagnosis.

Auditing the Synthetic Validation Trap

In preparation — pre-registration OSF-ready

- A downstream benchmark testing whether depression detectors trained on LLM-generated text generalize to real humans: 11 training conditions, three detector families (TF-IDF+LR, SBERT+LR, fine-tuned RoBERTa), six pre-registered hypotheses, hierarchical bootstrap ($M \times B = 100 \times 1000$), and a frozen 316-positive cohort with full SHA-256 data-integrity verification.

SELECTED RESEARCH PROJECTS — SYSTEMS & ENGINEERING

Pre-Registered LLM-Audit Pipeline (STLE → PGG → Downstream)

- End-to-end evaluation infrastructure in Python: LoRA persona generation, fine-tuned RoBERTa and SBERT/TF-IDF detectors, hierarchical bootstrap, calibration analysis (ECE, Brier), and purpose-built diversity/geometry instruments (TADA, FRFA, MAUVE, MMD).
- Reproducibility engineering: OSF pre-registration, seed-locked stratified splits, leakage-prevention protocols, SHA-256 integrity verification of frozen cohorts, resumable parquet caching, and anonymous review repositories.

TSS — White-Box Diagnostic Framework

- Multi-channel, interpretable probe separating lexical, morpho-syntactic, and psycholinguistic signals to test whether mental-health classifiers learn clinical patterns or shortcut features; includes an interventional masking suite for quasi-causal evidence.

TECHNICAL SKILLS

Machine Learning & LLMs: PyTorch; Hugging Face Transformers (RoBERTa / BERT / BioBERT / MentalBERT sequence classification, tokenization); parameter-efficient fine-tuning (LoRA / PEFT); sentence-transformers (SBERT); fine-tuning, linear probing, frozen-encoder training; activation steering.

Interpretability experiments: nnsight (activation capture & intervention), TransformerLens; hidden-state probing; sparse-autoencoder / feature-based analysis; classifying cognitive states from internal representations.

LLM & Synthetic-Data Evaluation: Distributional-fidelity auditing; MAUVE, MMD; purpose-built diversity / geometry instruments (TADA, FRFA); persona generation & auditing; calibration analysis (ECE, Brier); benchmark construction.

Statistics & Experimental Design: Bootstrap confidence intervals (incl. hierarchical $M \times B$ resampling), permutation tests, BH-FDR multiple-comparison control, Difference-in-Differences, ElasticNet regularization, Cohen's d, ARI; pre-registration; leakage-prevention and ablation design.

NLP Methods: Text classification; TF-IDF, character n-grams, morpho-syntactic & psycholinguistic features; NMF topic modeling; spaCy / scispaCy NER & entity normalization; UMLS; OCR-noise modeling; deterministic spell-correction; abstention & safety gating.

Programming & Tooling: Python (pandas, NumPy, scikit-learn), SQL, Git / GitHub; LaTeX; Node.js; Jupyter / Colab (incl. A100 GPU); reproducible pipelines with seed control, SHA-256 verification, and parquet caching; OSF.

Institutional Data & Governance: Oracle E-Business Suite, Power Query, PivotTables; audit-ready reporting, data lineage / traceability, and documented controls.

ACADEMIC SERVICE

- Program Committee Reviewer and Meta-Reviewer, ACL BioNLP 2026.

EDUCATION

M.Sc. in Artificial Intelligence — Asia Pacific University of Technology & Innovation (APU), Malaysia
Expected 2027

M.A. in Comparative Literature — Doha Institute for Graduate Studies, Qatar *2017*

B.Com. in Finance — Alexandria University, Egypt *2004*

Selected coursework: Python for Data Science & AI (IBM / Coursera, 2025); Linear Algebra for Machine Learning (DeepLearning.AI / Coursera, 2025); Finance & Quantitative Modeling (University of Pennsylvania / Coursera, 2022).

PROFESSIONAL BACKGROUND (CONDENSED)

Senior Accountant — Doha Institute for Graduate Studies, Doha, Qatar *2014 – present*

- Built ERP-to-analytics reporting workflows (Oracle EBS, Power Query, PivotTables) producing audit-ready management packs with traceable data lineage; established the founding finance department, chart-of-accounts architecture, and documented financial controls.
- *This decade of data-governance and auditability work directly informs my research emphasis on traceable, reproducible, and appropriately abstaining AI systems.*

Earlier: Head of Treasury / Accounts Manager, SAKR for International Investment & Cleopatra Ceramics Group, Egypt (2008–2013) — multi-currency treasury, forecasting, and trade-finance operations.

LITERARY & HUMANISTIC BACKGROUND

- Published Arabic novelist and literary researcher (three novels: *The Sniper*, *The Return from the South*, *Dream of Ifenza*); literary website at escapingplanetearth.blogspot.com. Work on political violence, memory, exile, and moral ambiguity strengthens sensitivity to discourse, ambiguity, and human meaning in NLP evaluation.

TARGET ROLES & AVAILABILITY

- Target roles: AI/NLP Researcher; LLM / Model Evaluation; AI Safety & Model Auditing; Computational Mental-Health & Biomedical NLP.
- Locations: Qatar / GCC / Egypt / USA / Canada / Remote. Notice period: 2 months.